

M49 — Clinical alarm sounds on the Nursing Station + shared audio asset library

Phase: Phase 7 follow-on (post-M48, operator-supplied assets) **Status:** **DONE** **Blocked by:** M26 (alarm bus), M27 (Nursing Station), M48 (threshold alarms) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

The operator dropped 15 production-ready clinical-alarm WAV files into a sibling folder (`~/Documents/Claude/Projects/Multipatient multi student simualtion/sounds/clinical_alarms/`) with a manifest naming each file by its clinical context (priority + metric). M49 imports them into the project as a shared static asset library and wires them into the Nursing Station's alarm board so each new alarm plays the right clinical-alarm tone — same as a real bedside monitor.

The 15 files cover:

File	Purpose
01	Bed exit alarm — patient-side warble
02	Call bell — patient room chime
03	Code blue — emergency response cue
04 / 08 / 12	ECG waveform alarm — HIGH / MEDIUM / LOW
05 / 09 / 13	Heart rate alarm — HIGH / MEDIUM / LOW
06 / 10 / 14	SpO ₂ alarm — HIGH / MEDIUM / LOW
07 / 11 / 15	Respiratory rate alarm — HIGH / MEDIUM / LOW

Severity → priority mapping:

Alarm severity	Audio priority bucket
critical	high
warning	medium
info (default)	low

2. Structure

Files added:

- `portal/static/sounds/clinical_alarms/*.wav` (15 files + `MANIFEST.txt`) — copied from the operator's drop folder.
- `portal/alarm_sounds.py` (new) — mapping module:
- `severity_to_priority(severity) → "high" / "medium" / "low"`
- `audio_url_for(alarm)` — picks the right WAV path based on `source, metric, kind, severity`. Returns `None` when no curated WAV matches (UI silently skips audio for that alarm).
- `annotate(alarms)` — mutates each alarm dict in-place to add `audio_url` + `audio_priority` fields. Returns the list.

Files touched:

- `portal/alarms.py` — `active_alarms(room)` now calls

`alarm_sounds.annotate(out)` before returning so every alarm in the response carries `audio_url` (or `None`).

- portal/static/nurse_station.js:

- `renderAlarmBoard(alarms)` calls a new `_playNewAlarmSounds`

helper before rendering. - `_playNewAlarmSounds(alarms)` plays each alarm's `audio_url` via a freshly-built `Audio()` element at volume 0.8. - A module-scoped `_seenAlarmIds` set dedupes: an alarm fires its WAV once per occurrence. When the alarm leaves the active list, its id is removed from the seen set — so a recurrence fires the sound again. - When the alarms list goes empty, the seen set is cleared. - The `.play()` Promise's `.catch()` swallows the autoplay- policy rejection silently (first user gesture unlocks audio).

No backend schema migration. No new dataclass field. The audio fields are computed at read time from data already on the alarm dict.

3. Uses

3.1 Alarm → sound mapping

Source	Metric / Kind	Severity	→ File
threshold	hr	critical	→ 05 (HR HIGH)
threshold	hr	warning	→ 09 (HR MEDIUM)
threshold	hr	info	→ 13 (HR LOW)
threshold	spo2	critical	→ 06 (SpO2 HIGH)
threshold	spo2	warning	→ 10
threshold	spo2	info	→ 14
threshold	rr	critical	→ 07
threshold	rr	warning	→ 11
threshold	rr	info	→ 15
threshold	rhythm	critical	→ 04 (ECG HIGH)
threshold	rhythm	warning	→ 08
threshold	rhythm	info	→ 12
scene	code.blue	critical	→ 03 (code blue)
scene	vitals.drop	<severity>	→ HR family
device	alarm.injected.call_bell	<severity>	→ 02 (call bell)
device	alarm.injected.bed_alarm	<severity>	→ 01 (bed exit)
device	alarm.injected.code_blue	<severity>	→ 03 (code blue)
device	pump.alarm / cabinet alarm		→ None (device's own audio)
*	unknown		→ None

The mapping is intentionally permissive — alarms that don't have a curated WAV get `audio_url=None` and the UI silently skips audio for them (the visual badge still flashes).

3.2 Dedupe semantics

The Nursing Station polls `/api/room/alarms` every 3 s. Without dedupe, an unresolved breach (e.g. SpO₂ still 84 → still alarming) would re-play the WAV on every tick — operator hears a continuous loop of the same sound.

`_seenAlarmIds` is a `Set<string>` of `alarm_id` values we've already played the sound for. On each tick:

1. Mark every alarm currently in the active list as `active`.
2. For each active alarm, if its id is NOT in `_seenAlarmIds` → play the WAV, add to seen set.
3. For each id in `_seenAlarmIds` that's NOT in `active` → drop it (so a re-occurrence later fires the sound again).
4. When the active list goes empty, clear the entire seen

set.

3.3 Browser autoplay policy

The first `Audio(...).play()` on a page may be silently rejected by Chrome/Safari if the user hasn't interacted with the page yet (autoplay policy). M49 catches the Promise rejection silently so the failure doesn't cascade. The first time the supervisor clicks ANYWHERE on the page (e.g. a Clear button), audio unlocks for the rest of the session.

4. Functions (exported API surface)

Symbol	Where	Purpose
<code>alarm_sounds.severity_to_priority(s)</code>	<code>portal/alarm_sounds.py</code>	Maps "critical/warning/info" → "high/medium/low".
<code>alarm_sounds.audio_url_for(alarm)</code>	same	Returns the WAV URL for an alarm dict, or None.
<code>alarm_sounds.annotate(alarms)</code>	same	In-place mutator that adds <code>audio_url</code> + <code>audio_priority</code> to every alarm.
<code>_playNewAlarmSounds(alarms)</code>	<code>portal/static/nurse_station.js</code>	Audio-dispatch + dedupe.

5. Limitations

- **No volume / mute control.** Volume is hard-coded to 0.8.

A future M50 could add per-room volume + a mute toggle on the Nursing Station settings card.

- **Sounds play only on the Nursing Station.** Per-Patient Console

could ALSO play them (it has its own alarm visibility via the M48 threshold-breach feed), but operator workflow today is one supervisor at the nurses station + instructor on the console (silent). If operators ask for sounds on the console too, the same `_playNewAlarmSounds` pattern drops in.

- **Pump/cabinet device alarms keep their existing audio.** Those

are played by the device tablet's own `engine.handle` event push via the device WS —

`alarm_sounds.audio_url_for` returns None for them so the nursing station doesn't duplicate the sound.

- **No sound for newly-bound device events** (e.g. a device

joined the room). Out of scope; we only sound the alarms.

- **Browser autoplay policy** may delay the very first alarm

sound until the supervisor clicks somewhere. After the first user gesture audio unlocks for the session.

Mitigation: the visual alarm badge always shows immediately regardless of audio.

- **The WAVs are large** (~176 KB each, 2.6 MB total). Acceptable

for a LAN install; if the operator deploys to a remote tablet the first load adds ~2.6 MB. Each WAV is cached by the browser after first play.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_clinical_alarm_sounds.py::test_clinical_alarm_wavs_shipped	All 15 WAV files + MANIFEST.txt present in portal/static/sounds/clinical_alarms/	PASS	2026-05-27
...::test_severity_to_priority_mapping	critical→high, warning→medium, info→low	PASS	2026-05-27
...::test_audio_url_for_threshold_hr_critical	HR breach @ critical → file 05	PASS	2026-05-27
...::test_audio_url_for_threshold_spo2_critical	SpO ₂ breach @ critical → file 06	PASS	2026-05-27
...::test_audio_url_for_threshold_rr_warning	RR warning → file 11 (medium)	PASS	2026-05-27
...::test_audio_url_for_threshold_rhythm_mapped_to_ecg_family	Dangerous rhythm → ECG family WAV	PASS	2026-05-27
...::test_audio_url_for_scene_code_blue	scene + code.blue → file 03	PASS	2026-05-27
...::test_audio_url_for_device_call_bell	device + call_bell → file 02	PASS	2026-05-27
...::test_audio_url_for_device_bed_alarm	device + bed_alarm → file 01	PASS	2026-05-27
...::test_audio_url_none_for_pump_alarm	Pump alarms get None (device owns audio)	PASS	2026-05-27
...::test_audio_url_none_for_unknown_alarm	Unknown shape → None, no crash	PASS	2026-05-27
...::test_annotate_adds_audio_fields_to_all_alarms	annotate mutates in-place; adds both audio fields	PASS	2026-05-27
...::test_room_alarms_response_carries_audio_url_per_alarm	/api/room/alarms JSON carries audio_url + audio_priority per alarm; SpO ₂ critical → file 06	PASS	2026-05-27
...::test_static_wav_route_serves_actual_file	The static handler serves the WAV bytes (RIFF header + audio/* content-type)	PASS	2026-05-27
...::test_nurse_station_js_plays_audio_url_on_new_alarm	JS dispatcher uses new Audio(url) + .play() driven by a.audio_url	PASS	2026-05-27
...::test_nurse_station_js_dedupes_by_alarm_id	_seenAlarmIds set + add/has/delete pattern present	PASS	2026-05-27
Full v7 suite	359 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-codre	Initial M49: copied 15 WAV files + MANIFEST into portal/static/sounds/clinical_alarms/; new portal/alarm_sounds.py with mapping helpers; alarms.py annotates each alarm; nurse_station.js plays + dedupes; 16 new tests	portal/static/sounds/clinical_alarms / (15 files + MANIFEST.txt), portal/alarm_sounds.py (new), portal/alarms.py, portal/static/nurse_station.js, tests /v7/test_clinical_alarm_sounds.py (new)

8. Open questions / known issues

- **Volume + mute control on the Nursing Station.** Today

hard-coded volume=0.8. A future M50 could add an operator- facing volume slider + a temporary mute toggle (mutes for N seconds, then auto-unmutes so a critical alarm isn't missed).

- **Sounds on the Per-Patient Console.** Same `audio_url` field

is now in the alarm response; an instructor console could also play sounds if operators want it. Today only the Nursing Station plays — the instructor's console is silent because the instructor is usually focused on chat/scene injection.

- **Per-encounter audio routing in a single browser.** If two

encounter consoles are open in two tabs, both will play sounds. Acceptable; matches real bedside monitor distribution where each station has its own speakers.

- **Asset compression.** The WAV files are uncompressed PCM.

Converting to MP3 or OGG would cut the asset payload by ~10×. Out of scope for v7.0 — the WAVs play instantly with no decode latency, which matters more for alarm response time than bandwidth.

- **Audio cue for ALARM CLEARED.** Currently silent on clear.

Future M50 could add a brief acknowledgement chime when the supervisor clears an alarm.

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